

Foreword

Math Research - A Marathon, an Obsession, and a Beautiful Picture

Professor Michael Hutchings (Class of 1989) is a mathematician and a professor at University of California, Berkeley. He attended Stuyvesant from 1986 to 1989, where he contributed to the Math Team and Math Survey, and Harvard College from 1989 to 1993. As an undergraduate student at Harvard College, Hutchings did a Research Experience for Undergraduates (REU) project with Frank Morgan at Williams College that launched his interest in the mathematics of soap bubbles. After postdoctoral and visiting positions at Stanford University, the Max Planck Institute for Mathematics in Bonn, Germany, and the Institute for Advanced Study in Princeton, New Jersey, he joined the University of California, Berkeley faculty in 2001. He is currently the Chair of the Math Department at University of California, Berkeley and is known for the development of embedded contact homology and for proving the double bubble conjecture on the shape of two-chambered soap bubbles.

This foreword is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

My math journey started when I entered Stuyvesant as a sophomore after I moved to New York City. One day, I just wandered into the Math Team, which had practice every day at 8:00 AM, where they were doing a practice test that day. The captain was Elizabeth Wilmer, who also later became a math professor, and the coach was Mr. Geller. They, along with some of the other older students in the Math Team, encouraged me and were essential mentors and teachers for me. A lot of what made Stuyvesant a good school for me was the fellow students. The Math Team connected into this whole circle of people and groups and activities. I went to the Hampshire College Summer Studies in Mathematics (HCSSiM), which I wouldn't have known about if I hadn't learned about it from the Math Team. Before that summer camp, I wasn't sure that I wanted to be a mathematician because I didn't really have much of an idea of what a mathematician did. I initially

thought a mathematician would compile tables of integrals, or try to compute more digits of Pi, which didn't sound terribly interesting to me. HCSSiM introduced more of what math research is like, which is completely different from competitions. I got at least a little glimpse of how there's this whole huge world of interesting math problems and cool, creative ideas that people are working on. I also wrote some articles for the *Math Survey* back then about tiling with quadrilaterals and how to do mental arithmetic quickly.

When I started my career in math, I had a very skewed view of things because the only really interesting math available to me was the Math Team. Math competitions were all about solving problems quickly. You have to find the clever trick, or maybe even read the mind of the person who wrote the question. It's like a sprint, but research is more like a marathon. It could take years to solve a problem. Collaboration is extremely important. Research is more about being persistent and working on complicated puzzles over a long period of time and collaborating and exchanging ideas with other people. You share ideas with other people because maybe I'm an expert on one thing but somebody else is an expert on something else, and by combining our knowledge we can do something really cool. I find that in research, if you're in the same room with your collaborators working on a problem together, it's sometimes really magical and you can do things which you never anticipated would be possible. When you finally figure the problem out, it's very satisfying. You usually find some beautiful picture.

As for the subject matter, my research area is in geometry and topology. This has little resemblance to the geometry that one usually sees in high school and on math competitions. The latter tends to focus on Euclidean geometry, and somewhat obscure questions about triangles and circles. I was actually never very good at that in math competitions. What modern geometers do is much more interesting. It's about curved spaces in three or four dimensions or higher, such as the shape of the whole universe, black holes, soap bubbles, or knots, to name a few. There are also some cool things about being a mathematician - there's tons of travel to participate in conferences and work with collaborators, which I had no idea about. I've spent a total of years of my life in other countries for work, especially in Germany, Brazil, and East Asia. I'm very lucky that I get to work on cool puzzles as part of my job.

The first time that I really failed badly at something was in the middle of graduate school. I spent a year trying to solve a certain math problem, and I thought I had a good idea for how to solve this problem. I thought I figured it out and I worked hard to write a research paper explaining my solution to the problem. Then it turned out, a couple of days after I finished the paper, I discovered there is a serious error in there and no apparent way to fix it. So this whole year of work was a total failure. This happens of course; it's a part of life as you try hard things. You're going to have to fail some of the time. The younger

you can have this experience and get over it the better because it gets harder. In my case, I was very discouraged and I was considering quitting graduate school and getting a computer programming job. But I pulled myself together and found some other good stuff to work on with the help of my advisor, which eventually led to my PhD thesis.

Working on a problem for years requires a certain kind of personality. You have to be obsessed. Now I should clarify that it's not that you pick one problem. Usually, you have various problems that you're working on. It's hard to isolate good questions that you can hope to answer. Some of them might be impossible, or maybe you'll never figure them out and somebody else will later. And that's okay. You don't have to solve everything, and often you also don't really know what exactly the question is. In a math homework assignment or on a math contest, you're given a question and you answer it if you can, but in math research, many times you don't even know what the question is. It also takes a really long time to write your research papers. If all I had to do was sit and write my research papers all day, I would be very happy. Unfortunately, this is unfeasible.

In terms of my research, my biggest accomplishment is the development of a theory in three- and four-dimensional geometry and topology called "embedded contact homology". In retrospect, this was a big gamble, because I worked for years on developing this theory, without knowing in advance whether it would be successful or have any value. I wasn't really thinking of it like that, though; I was just working on it because I was interested in it. In any case, it turned out to be successful and useful for some problems in geometry, specifically symplectic embeddings (a kind of packing problem), and dynamics of Reeb vector fields (roughly speaking, understanding behavior of physical systems that repeats in time). Aside from doing research, I have taught many thousands of students in my classes over my whole career. A lot of the students probably found my class boring, and maybe didn't get that much out of it. But just given such a large number of students, if I helped some of them learn to think more logically about quantitative questions, that might make more of an impact on the world than any of my research.

During the pandemic, a few things in regards to math research have arguably gotten better because there are research seminars and conferences that are now held online. Anybody from anywhere in the world can come and learn. In regards to the future of mathematics, there is some talk about proof assistants, where you have some software where you can write your mathematical proof with computer code, and the proof assistant can verify that your proof is correct. The proof assistant can even fill in some small details, so you don't necessarily have to write down every single step. However, I'm not sure that's going to happen that much with the kind of geometry that I do; it's very difficult to translate geometrical reasoning into computer code. And even if you do, there's a problem making sure that the interpretation of your computer code is actually what you were trying to say.

So in other words, a lot of the difficulty gets shifted onto making sure that your definitions are really defining what you mean to be defining. So while that's an interesting development, I think it will be a while before it has a big impact on the practice of mathematics. And even if a computer can check that your proof is correct, it seems unlikely that computers can replace the human intuition that led to the proof in the first place.

There is also room for more computer experimentation in mathematics, which seems to be underutilized at least in the kind of geometry that I work on. There will most likely be more potential for students to go into mathematics with good computer programming skills that can be used to do experiments, to test mathematical conjectures.

My advice for those of you aspiring to go into math would be to follow what you're interested in, talk to other people as much as you can, and learn from other people. In the next 10 or 15 years, what being a mathematician means most likely won't change too much. Of course, there's very rapid progress in mathematics - new fields of mathematics emerge, new things become possible, and mathematical advances that would have been inconceivable 10 or 20 years previously become possible. I can't predict what's going to happen in terms of the content of mathematics. In my own specialty, there are some famous conjectures; personally I am most obsessed with one called the Viterbo conjecture. I try to make little bits of progress related to some of these conjectures. So maybe we can solve some of these, and I am certainly very excited to see what will happen. But in many ways, the practice of mathematics research will probably stay the same. There's not that much to it. It's reading papers, thinking, talking to people, and writing papers. And I think we're going to keep doing that.

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December 2020